

Gabriel Cruz

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science and Mathematics | GPA: 3.2

Aug 2024 – May 2028

- Threads: Intelligence and Systems & Architecture
- Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Computer Architecture, Discrete Mathematics, Intro to AI, Probability & Statistics, Linear Algebra, Machine Learning, Differential Equations, Calculus I-III, Physics I-II
- Clubs: Trading @ GT, Chess Club

EXPERIENCE

Johns Hopkins Applied Physics Laboratory

Laurel, MD

Software Engineering Intern — Python, NumPy, GNU Radio, Arduino, Plotly

June 2023 – May 2024

- Simulated 2.45 GHz and 5.8 GHz ISM band signals in GNU Radio, reducing energy rebound by 30%.
- Developed predictive Free Space Path Loss model, improving wireless system design accuracy.
- Prototyped low-cost 2.45 GHz closed-loop WPT system with adaptive phase and impedance tuning.
- Built automated plant health monitoring pipeline (Arduino + Python + Plotly), achieving 99.1% detection accuracy.

PROJECTS

Roadrunner Mock Interviewer | *React, FastAPI, Python, NumPy, Supabase, PostgreSQL*

- Built a full-stack AI mock interview platform with a React frontend and FastAPI backend, generating adaptive interview questions and feedback from a candidate's resume and target job description via the Gemini API.
- Engineered a logistic regression resume-classifier from scratch in NumPy using 61 hand-crafted features (skills, titles, experience signals), achieving 94.8% accuracy and 0.996 ROC-AUC on 44K+ labeled resumes.
- Integrated the ElevenLabs API to stream real-time text-to-speech, giving the mock interviewer a live conversational voice and modeling user auth/storage in PostgreSQL via Supabase.

Breast Cancer Survival Prediction Pipeline | *Python, scikit-survival, XGBoost, SHAP, Pandas, NumPy*

- Built a survival analysis pipeline on the METABRIC breast cancer dataset (1,903 patients, 629 clinical and genomic features), boosting concordance index from 0.60 to 0.75 with a Random Survival Forest over a Cox Proportional Hazards baseline.
- Built a leakage-free preprocessing pipeline with KNN imputation, variance filtering, and stratified train/validation/test splitting, then calibrated survival probabilities via isotonic regression to under 1% prediction error.
- Applied SHAP and permutation importance to surface top prognostic biomarkers (HSD17B11, Nottingham Prognostic Index, age at diagnosis), identifying genes with hazard ratios up to 5x baseline risk.

Cache-Blocked Matrix Multiplication | *C++*

- Implemented a custom `Matrix<T>` class with flat row-major memory layout, then built a naive triple-loop multiplication and a cache-blocked (tiled) version to reduce cache misses on large matrices.
- Benchmarked both implementations on 512x512 matrices using a custom GFLOPs/sec harness, achieving a 6.3x speedup and improving utilization from 3.0% to 19.1% of theoretical peak FLOPs.
- Identified the naive version's bottleneck as poor cache reuse rather than algorithmic complexity, since both implementations remain $O(n^3)$, motivating the blocking strategy.

TECHNICAL SKILLS

Languages: Assembly, C++, Python, Java

Frameworks: Django, JUnit

Developer Tools: Git, GitLab, Github, VS Code, Visual Studio, PyCharm, IntelliJ, Eclipse

Libraries: Pandas, NumPy, PyAlgoTrade, SciPy, Scikit-learn, XGBoost, TensorFlow, Jupyter Notebook

Hardware/Lab Tools: GNU Radio, Fusion 360, Oscilloscope, Multimeter, FPGA